1. Suction type
2. Blower type
3. Electrical type

7.8 ANTI-FREEZING SOLUTIONS

During cold climate conditions and at hill stations water generally gets frozen in the water cooling system. Due to this radiator core, water jacket and rubber hoses in the cooling system may get damaged. In addition more power is needed to start the engine. To overcome these difficulties chemical agents are added to the water to prevent it from freezing. These agents are called as anti freezing solutions. Some of the anti freeze solutions used in automobile are,

1. Methanol
2. Methyl alcohol

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Cooling System Capacity ▼	ANTIFREEZE PROTECTION CHART (°F) — Quarts Required for Low-Temperature Protection ►									
QUARTS	3	4	5	6	7	8	9	10	11	12
9	5°	-15°	-43°	-76°						
10	10°	-4°	-26°	-54°	-76°					
11	12°	0°	-14°	-40°	-60°					
12	14°	5°	-8°	-28°	-51°					
13	15°	8°	0°	-14°	-44°	-61°	-76°			
14	17°	10°	3°	-8°	-28°	-44°	-60°			
15	18°	12°	5°	-4°	-14°	-35°	-54°	-76°		
16	19°	14°	9°	1°	-9°	-28°	-44°	-60°		
17	20°	16°	11°	3°	-2°	-15°	-31°	-60°		
18	21°	17°	12°	5°	0°	-13°	-26°	-33°	-53°	-76°
19		18°	13°	7°	2°	-10°	-20°	-32°	-50°	-60°
20			14°	9°	3°	-6°	-15°	-26°	-33°	-54°
21			15°	12°	5°	0°	-10°	-22°	-32°	-51°
22			16°	13°	8°	3°	-5°	-10°	-26°	-33°
23			17°	13°	10°	4°	-3°	-8°	-22°	-32°



Figure 7.8 Anti-freezing solutions

3. Glycerin
4. Alcohol
5. Ethylene glycol

The above chemical components must have the requirements such as easily soluble in water, should withstand very high temperature, non-corrosiveness and should not deposit on the radiator core and rubber hose.

7.9 MAINTENANCE OF COOLING SYSTEM

The following are the check lists for the proper maintenance of the cooling systems

1. If there is any blockage occurs inside the radiator, water tubes and water jackets they have to be cleaned by flushing process.
2. The fan belt has to be replaced if it is exhausted or cracked.
3. The shape of fan blade has to be checked whether it is ok or not.
4. The cells must be straightened in the radiator if they are blended.
5. If there are any unwanted materials or insects nest are present in the radiator blades, they must be cleaned by using compressed air.
6. If there is any damage or hole occurs in the radiator tubes (rubber), it has to be replaced.
7. Fine tight must be given to the radiators tube clips.
8. Leakage in the radiator pipe, jar, tank, water pump, thermostat valve and drain gauge must be checked and leak proof has to be ensured.

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